

# A tutorial on cjr: syntax, semantics, and separation logic

Mechanized development in **theories/**

## Abstract

CJR (Cangjie Radical) is the performance-parity core of a Cangjie-like language: unboxed structs, word-addressed **CPointer** blocks, and final classes, with **func**, **if**, and **while**. This note is a reading guide to the Coq/Iris mechanization. Every concept is given twice: as the definition that is actually checked, and as ordinary mathematics with the same content. Every program-logic rule is followed by what it is for. The note ends by walking through the eight examples that exercise the rules in order.

## 1 What this development is

CJR is sequential, monomorphic, and first-order, with code pointers so that a vtable can be written by hand. There are no generics, no inheritance, no **enum/match**, and no concurrency. Those features of a larger source language are treated as sugar that must lower into this core at the same time and space cost. This development defines the core. It does not define the lowering, a garbage collector, or byte-level layout.

Safety is not a type system. The grammar **ty** records the types one intends to use, but the operational semantics is untyped. A program is safe when a weakest-precondition proof exists: adequacy then says that, from the empty machine, execution does not get stuck.

The Coq sources live under **theories/**. The language is an Iris evaluation-context language, so the frame rule and the bind rule are Iris's, not a separate Hoare calculus.

## 2 Syntax

### 2.1 Locations, identifiers, and types

A location is an integer. Stack slots, raw words, and objects use distinct wrappers so that the four ghost maps of the program logic cannot be confused.

```
Definition loc : Type := Z.
Inductive stack_id := StackId (z : Z).
Inductive raw_id := RawId (z : Z).
Inductive obj_id := ObjId (z : Z).

Inductive ty :=
  | TyInt | TyFloat | TyBool | TyUnit
  | TyStruct (ts : list ty)
  | TyPtr (t : ty)
  | TyObj (nfields : nat)
  | TyFun (t1 t2 : ty).
```

Informally, locations are  $\mathbb{Z}$ , and the three identifier sorts are the tagged copies  $\text{StackId}(\ell)$ ,  $\text{RawId}(\ell)$ , and  $\text{ObjId}(\ell)$ . The type grammar is

$$\tau ::= \text{Int64} \mid \text{Float64} \mid \text{Bool} \mid \text{Unit} \mid \text{struct } \bar{\tau} \mid \text{CPointer } \tau \mid \text{class}_n \mid \tau \rightarrow \tau.$$

$\text{class}_n$  means a final class with  $n$  fields. Nothing in the semantics consults  $\tau$ . The type is documentation of the fragment; the theorems that follow do not assume it.

## 2.2 Values and expressions

A binder is an optional name: `None` is an ignored parameter. Literals cover integers, a float payload (no float operations are defined), booleans, unit, raw pointers, null, object references, and stack addresses. Values are literals, recursive closures, and unboxed struct tuples. Expressions are the core forms, plus administrative forms (`Pop`, `StackLoad`, `StackAssign`, `StackFieldStore`) introduced when a mutable variable is allocated.

```

Definition binder := option string.

Inductive base_lit :=
  | LitInt (n : Z) | LitFloat (n : Z) | LitBool (b : bool)
  | LitUnit | LitPtr (l : loc) | LitNull
  | LitObj (o : loc) | LitStack (l : loc).

Inductive val :=
  | LitV (l : base_lit)
  | RecV (f x : binder) (e : expr)
  | StructV (vs : list val)
with expr :=
  | Val (v : val) | Var (x : string)
  | Rec (f x : binder) (e : expr)
  | App (e1 e2 : expr)
  | Let (x : binder) (e1 e2 : expr)
  | VarBind (x : string) (e1 e2 : expr)
  | Assign (x : string) (e : expr)
  | Pop (l : stack_id) (e : expr)
  | StackLoad (e : expr)
  | StackAssign (e1 e2 : expr)
  | StackFieldStore (e1 : expr) (i : nat) (e2 : expr)
  | If (e0 e1 e2 : expr)
  | While (e1 e2 : expr)
  | Seq (e1 e2 : expr)
  | BinOp (op : bin_op) (e1 e2 : expr)
  | Alloc (e : expr) | Free (e : expr)
  | Load (e : expr) | Store (e1 e2 : expr)
  | Offset (e1 e2 : expr)
  | Struct (vs : list val) (es : list expr)
  | StructLoad (e : expr) (i : nat)
  | StructStore (x : string) (i : nat) (e : expr)
  | New (vs : list val) (es : list expr)
  | FieldLoad (e : expr) (i : nat)
  | FieldStore (e1 : expr) (i : nat) (e2 : expr).

```

In mathematics we write values

$$v ::= n \mid f \mid b \mid () \mid \text{ptr}(\ell) \mid \text{null} \mid \text{obj}(\ell) \mid \text{stk}(\ell) \mid \text{rec } f x. e \mid \langle v_0, \dots, v_{k-1} \rangle$$

and use the expression forms by the same names as the constructors. **BinOp** ranges over  $+$ ,  $-$ ,  $\leq$ ,  $=$ , and boolean conjunction. Integer operations act on **LitInt**. Conjunction acts on booleans. Anything else makes **bin\_op\_eval** return **None**, and the machine gets stuck.

Structs and objects are built by a list zipper. **Struct**  $\bar{v} \bar{e}$  has already evaluated the values  $\bar{v}$  and still has the expressions  $\bar{e}$  to evaluate, left to right. **New** is the same zipper for a final class. When the expression list is empty, the zipper is done.

**Val** and **to\_val** identify which expressions are finished:

```
Definition of_val (v : val) : expr := Val v.
Definition to_val (e : expr) : option val :=
  match e with Val v => Some v | _ => None end.
```

So the values of the language, as Iris sees them, are exactly the expressions **Val**  $v$ .

## 2.3 Two kinds of binding

Immutable **let** and function parameters are substitution. Mutable **var** is not. A **var** is given a fresh stack slot, and every use or assignment of that name is rewritten into an operation on the slot. A bare struct value is still copied: mutation in place happens only through a **var** slot or through a stored field.

```
Definition subst_binder (b : binder) (v : val) (e : expr) : expr :=
  match b with None => e | Some x => subst x v e end.
```

The function **subst**  $x v e$  replaces free occurrences of  $x$  by the value  $v$ , and it does not descend under a binder that captures  $x$ . On a value, literals are unchanged and the body of a closure is substituted only when  $x$  is not the function name or the parameter.

The mutable variant **subst\_var**  $x \ell e$  does a different job. A read **Var**  $x$  becomes **StackLoad**(**stk**( $\ell$ )). An assignment **Assign**  $x e$  becomes **StackAssign**(**stk**( $\ell$ ),  $e'$ ). A field write **StructStore**  $x i e$  becomes **StackFieldStore**(**stk**( $\ell$ ),  $i$ ,  $e'$ ). Other occurrences of  $x$  are rewritten recursively, again respecting capture. The slot address is a value, so later stack operations do not look up a string environment.

This split is the whole model of “locals do not need a points-to fact until they are mutable.” An immutable binding disappears into the syntax. A mutable binding is a cell, and the cell has an owner.

## 3 Machine state

### 3.1 Three regions, one counter

The machine is one record. Locals and unboxed structs that are stored in **var** slots live in the stack map. **CPointer** words live in the raw map, and each allocated block records its length in a second map on the same addresses. Final class fields live in a map from (**ObjId**( $\ell$ ),  $i$ ) to a value. A single counter **next** hands out fresh addresses for all three regions.

```
Record state := MkState {
  cjr_stack : gmap stack_id val;
  cjr_raw   : gmap raw_id val;
```

```

cjr_blocks : gmap raw_id nat;
cjr_obj     : gmap (obj_id * nat) val;
cjr_next    : Z
}.
Definition state_init : state := inhabitant. (* all maps empty, next = 0 *)

```

Write a state as

$$\sigma = (\sigma_s, \sigma_r, \sigma_b, \sigma_o, \text{next}(\sigma)).$$

The initial state  $\sigma_0$  is  $(\emptyset, \emptyset, \emptyset, \emptyset, 0)$ .

Indexing a pointer is arithmetic on the address, one word per element:

$$\text{ptr}(\ell) + i = \text{ptr}(\ell + i).$$

There is no byte offset and no padding. A block of length  $n$  at base  $\ell$  occupies the words  $\ell, \ell + 1, \dots, \ell + n - 1$ . Blocks are freed explicitly. Class objects are not freed in this development: there is no `delete` for them, and there is no reachability argument.

Null is a pointer value, not an address in  $\sigma_r$ . Loading or storing through null has no rule, so it is stuck. An out-of-bounds or use-after-free access is stuck for the same reason: the lookup premise of load, store, or free fails.

### 3.2 Well-formed states

Not every 5-tuple is a state the allocator is allowed to produce. Well-formedness says that every address already in use lies strictly below the counter, that every block's words are present, and that live blocks do not overlap.

```

Definition stack_below m n :=
  forall l, StackId l in dom m -> (l < n)%Z.
(* raw_below, block_below, obj_below are the same idea. *)

Definition block_cover blocks raw next :=
  forall l n, blocks !! RawId l = Some n ->
    (l + Z.of_nat n <= next)%Z /\
    forall i, i < n -> is_Some (raw !! RawId (l + Z.of_nat i)).

Definition blocks_disjoint blocks :=
  forall l1 n1 l2 n2,
    blocks !! RawId l1 = Some n1 ->
    blocks !! RawId l2 = Some n2 ->
    l1 <> l2 ->
    (l1 + Z.of_nat n1 <= l2 /\ l2 + Z.of_nat n2 <= l1)%Z.

Record state_wf (s : state) : Prop := {
  wf_stack : stack_below (cjr_stack s) (cjr_next s);
  wf_raw    : raw_below (cjr_raw s) (cjr_next s);
  wf_block  : block_below (cjr_blocks s) (cjr_next s);
  wf_obj    : obj_below (cjr_obj s) (cjr_next s);
  wf_cover  : block_cover (cjr_blocks s) (cjr_raw s) (cjr_next s);
  wf_disj   : blocks_disjoint (cjr_blocks s)
}.

```

In mathematics,  $\text{wf}(\sigma)$  is the conjunction of:

$\forall \ell. \text{StackId}(\ell) \in \text{dom}(\sigma_s) \Rightarrow \ell < \text{next}(\sigma)$ ,  
 and the same for raw addresses, block bases, and object addresses,  
 $\forall \ell, n. \sigma_b(\text{RawId}(\ell)) = \text{Some}(n) \Rightarrow \ell + n \leq \text{next}(\sigma) \wedge \forall i < n. \sigma_r(\text{RawId}(\ell + i)) \neq \text{None}$ ,  
 distinct live blocks have disjoint integer intervals.

The freshness lemmas are immediate from the first four clauses. If  $\text{wf}(\sigma)$  and  $\ell = \text{next}(\sigma)$ , then  $\ell$  is absent from the stack, from the raw heap, from the block map, and, at every field index, from the object heap. Allocation is therefore the insertion of a key that is not already present.

The initial state is well formed: **state\_init\_wf**. Each successful head step that the program logic uses is proved to preserve **wf**: stack allocation, deletion, and update; raw store; one-word allocation and freeing; class allocation; and field update.

One-word allocation is worth spelling out, because it is the case the logic rule talks about. From base  $\ell = \text{next}(\sigma)$ ,

$$\begin{aligned}\sigma'_r &= \{\text{RawId}(\ell) \mapsto 0\} \cup \sigma_r, \\ \sigma'_b &= \sigma_b[\text{RawId}(\ell) \mapsto 1], \\ \text{next}' &= \ell + 1.\end{aligned}$$

The new word is in range, the new block is covered, and disjointness holds because the new interval  $[\ell, \ell + 1)$  starts at the old counter, past every previous address. Freeing that block deletes the length and the word. Other blocks keep their cells precisely because their intervals were disjoint from  $[\ell, \ell + 1)$ .

Class allocation at  $o = \text{next}(\sigma)$  inserts the fields  $(\text{ObjId}(o), i) \mapsto v_i$  for a list  $v_0, \dots, v_{k-1}$  and advances the counter by one. Those keys are fresh because every old object address is below  $o$ .

## 4 Small-step semantics

### 4.1 Head reduction

A head step is a relation

$$(e, \sigma) \longrightarrow (e', \sigma')$$

with no observations and no forked threads. Both are hard-wired: **observation** is the empty type, and every rule produces an empty list of threads. The language is deterministic on the nose.

**Pure computation.** These rules do not read or write the heaps.

$$\begin{array}{ll}
\text{rec } f \ x. e \longrightarrow \text{rec } f \ x. e & \text{(as a value)} \\
(\text{rec } f \ x. e) \ v \longrightarrow e[x \mapsto v][f \mapsto \text{rec } f \ x. e] & \\
\text{let } x = \text{vine} \longrightarrow e[x \mapsto v] & \\
\text{if true then } e_1 \text{ else } e_2 \longrightarrow e_1 & \\
\text{if false then } e_1 \text{ else } e_2 \longrightarrow e_2 & \\
\text{while } e_1 \text{ do } e_2 \longrightarrow \text{if } e_1 \text{ then } (e_2; \text{while } e_1 \text{ do } e_2) \text{ else } () & \\
v; e \longrightarrow e & \\
v_1 \oplus v_2 \longrightarrow v & \text{when } \llbracket \oplus \rrbracket(v_1, v_2) = \text{Some}(v) \\
\text{ptr}(\ell) + i \longrightarrow \text{ptr}(\ell + i) & \\
\langle \bar{v}, v :: \bar{e} \rangle \longrightarrow \langle \bar{v} ++ [v], \bar{e} \rangle & \\
\langle \bar{v} \rangle \longrightarrow \langle v_0, \dots \rangle & \text{(the struct value)} \\
\langle \bar{v} \rangle . i \longrightarrow v & \text{when } v \text{ is the } i\text{th component.}
\end{array}$$

The **while** rule duplicates the loop. It is a head step, not an evaluation context, so the copies are syntax, not a shared mutable closure. The condition and the body are evaluated later, by ordinary contexts.

**Stack.** Let  $\ell = \text{next}(\sigma)$ .

$$\begin{array}{ll}
\text{var } x = v \text{ in } e \longrightarrow \text{pop}_\ell(e\{x \mapsto \ell\}) & \text{and } \sigma_s(\text{StackId}(\ell)) := v, \text{ next} := \ell + 1 \\
\text{pop}_\ell v \longrightarrow v & \text{and delete the slot} \\
\text{stk}(\ell) \longrightarrow v & \text{if the slot holds } v \\
\text{stk}(\ell) := v \longrightarrow () & \text{if the slot exists; it then holds } v \\
\text{stk}(\ell).i := v \longrightarrow () & \text{if the slot holds a struct and index } i \text{ is in range.}
\end{array}$$

The last rule replaces component  $i$  with **set\_nth** and writes the struct back into the same slot. No raw or object address is allocated. That is why a stack update can be framed against a raw points-to fact: the two maps are different components of  $\sigma$ .

**Raw heap.** For  $n > 0$ , allocation of  $n$  words at  $\ell = \text{next}(\sigma)$  returns  $\text{ptr}(\ell)$ , records length  $Z.\text{to\_nat}(n)$ , zeroes the words, and sets the counter to  $\ell + n$ . The proved program-logic rule specialises this to  $n = 1$ .

$$\begin{array}{ll}
\text{free}(\text{ptr}(\ell)) \longrightarrow () & \text{if a block length is recorded at } \ell; \text{ that block's words are deleted} \\
!\text{ptr}(\ell) \longrightarrow v & \text{if the word holds } v \\
\text{ptr}(\ell) := v \longrightarrow () & \text{if the word exists; it then holds } v.
\end{array}$$

Free deletes whatever length is stored. The one-word reasoning rule asks separately for the length fact 1, so the client of that rule cannot free a longer block by mistake.

**Objects.** The object zipper steps like the struct zipper. When it finishes, at  $o = \text{next}(\sigma)$ ,

$$\text{new}(v_0, \dots, v_{k-1}) \longrightarrow \text{obj}(o)$$

and the fields  $(o, i) \mapsto v_i$  are added. Field load and field store are lookup and update at  $(\text{ObjId}(o), i)$ . Store requires the field to exist already. A missing field, like a missing word, is stuck.

## 4.2 Evaluation contexts

Head steps fire only at the root. The context items say where a subexpression may be reduced. Each item has one hole, and the hole is the expression currently being evaluated. Arguments are evaluated left to right. `if` evaluates only the condition. `while` does not evaluate under itself: the unfolding rule is the head step above.

Representative items, written as contexts:

$$\begin{aligned} &E[\text{let } x = [] \text{ in } e], E[\text{var } x = [] \text{ in } e], E[\text{pop}_\ell []], \\ &E[[] e], E[v []], E[\text{if } [] \text{ then } e_1 \text{ else } e_2], E[[]; e], \\ &E[[] \oplus e], E[v \oplus []], E[! []], E[\text{free } []], E[[] := e], E[v := []]. \end{aligned}$$

Struct and `new` items evaluate the next expression in the zipper. The Coq constructor list is `ectx_item`, and `fill_item` implements the grammar.

Iris then builds the language in the standard way.

```
Lemma cjr_ectxi_mixin : EctxiLanguageMixin of_val to_val fill_item base_step.
Canonical Structure cjr_ectxi_lang := EctxiLanguage cjr_ectxi_mixin.
Canonical Structure cjr_ectx_lang := EctxLanguageOfEctxi cjr_ectxi_lang.
Canonical Structure cjr_lang := LanguageOfEctx cjr_ectx_lang.
```

A configuration steps by finding a context  $K$  and a head redex:

$$K[e] \longrightarrow K[e'] \quad \text{when} \quad (e, \sigma) \longrightarrow (e', \sigma').$$

Filling a non-value never produces a value, a value never head-reduces, and a head reduction of a filled term implies that the hole was already a value. Those are the mixin obligations. They are what let Iris treat  $K$  as a program context in the bind rule.

## 4.3 Determinism

```
Lemma base_step_det e s k e' s' t k2 e2 s2 t2 :
  base_step e s k e' s' t ->
  base_step e s k2 e2 s2 t2 ->
  k = k2 /\ e' = e2 /\ s' = s2 /\ t = t2.
```

**Theorem 1** (Determinism of head reduction). *If  $(e, \sigma) \longrightarrow (e_1, \sigma_1)$  and  $(e, \sigma) \longrightarrow (e_2, \sigma_2)$ , then  $e_1 = e_2$  and  $\sigma_1 = \sigma_2$ . Observations and forked threads, which are always empty, agree as well.*

The proof is inversion of both derivations. Where a rule reads a map or calls  $[\![\oplus]\!]$ , the two lookups are the same, so the two results are the same. There is no scheduler and no choice of fresh address other than  $\text{next}(\sigma)$ .

## 5 Separation logic

### 5.1 What is owned

The program logic does not talk about the whole machine. It talks about the cells a piece of code is allowed to use. Ownership is a ghost map, one authoritative map per region, with a points-to fact for each cell. Full ownership is the fraction 1.

```

Definition stack_mapsto (l : stack_id) (v : val) : iProp S :=
  ghost_map_elem stack_name l (DfracOwn 1) v.
Definition raw_mapsto (l : raw_id) (v : val) : iProp S :=
  ghost_map_elem raw_name l (DfracOwn 1) v.
Definition block_mapsto (l : raw_id) (n : nat) : iProp S :=
  ghost_map_elem block_name l (DfracOwn 1) n.
Definition obj_mapsto (o : obj_id) (i : nat) (v : val) : iProp S :=
  ghost_map_elem obj_name (o, i) (DfracOwn 1) v.

```

The notations in the sources are  $\ell \mapsto_s v$ ,  $\ell \mapsto_r v$ ,  $\ell \mapsto_b n$ , and  $o \mapsto_o [i]v$ . Four maps are used, rather than a single heap, because a stack slot, a raw word, a block length, and a class field are different kinds of cell. A stack proof cannot accidentally be applied to a pointer.

The state interpretation ties the ghost maps to the concrete machine and remembers well-formedness:

```

Definition cjr_state_interp (s : state) : iProp S :=
  ghost_map_auth stack_name 1 (cjr_stack s) *
  ghost_map_auth raw_name 1 (cjr_raw s) *
  ghost_map_auth block_name 1 (cjr_blocks s) *
  ghost_map_auth obj_name 1 (cjr_obj s) *
  pure (state_wf s).

```

At every step, Iris owns  $\text{StateInterp}(\sigma)$ . A points-to fact is a fragment of one of those authoritative maps. Updating a cell is updating both the fragment and the matching entry of  $\sigma$ , and the wf conjunct is re-established by the lemma for that step.

## 5.2 Weakest precondition, in this instantiation

The Iris instance is the standard one for a deterministic language with no thread pool of interest:

$$\begin{aligned}
\text{stateInterp}(\sigma) &= \text{cjr\_state\_interp}(\sigma), \\
\text{forkPost}(\_) &= \top, \\
\text{numLaterPerStep}(\_) &= 0.
\end{aligned}$$

The weakest precondition  $\text{wpe } \{\Phi\}$  is Iris's. For this tutorial the reading is:

**Definition 2** (Specification).  *$\text{wpe } \{\Phi\}$  means: for every well-formed extension of the current ghost ownership, executing  $e$  neither gets stuck nor forks, and if it terminates with a value  $v$  then  $\Phi(v)$  holds. Partial correctness is enough: a diverging loop satisfies a weakest precondition whose postcondition is never reached.*

The later  $\triangleright$  and the credit  $\mathcal{L}1$  appearing in the rules are Iris's accounting for steps, in the mode `HasLc`. When a rule says

$$\ell \mapsto_r v \multimap (\mathcal{L}1 \multimap \ell \mapsto_r v \multimap \Phi(v)) \vdash \text{wp } (!\ell) \{\Phi\},$$

the intended content is: if you own the cell, you may load it; you get the same cell back, together with the value that was there. The later and the credit are discharged by the step. In the example proofs they are thrown away with an introduction `iIntros " > _"`. They do not change the memory reasoning.

### 5.3 Structural rules

These three rules are why the logic is a separation logic rather than a Hoare logic with an explicit frame side-condition. Bind and the pure-step lemma are proved in `primitive_laws.v` by appealing to Iris. The frame rule is Iris’s own `wp_frame_1`; this development does not re-prove it.

```
Lemma wp_bind K e Phi :
  wp e (fun v => wp (fill K (Val v)) Phi) |- wp (fill K e) Phi.
```

**Lemma 3** (Bind). *If  $K$  is an evaluation context, then*

$$\text{wp } e \{v. \text{wp } K[v] \{\Phi\}\} \vdash \text{wp } K[e] \{\Phi\}.$$

*Intuition.* To prove a specification of a term with a hole, prove a specification of the hole, and allow the postcondition of the hole to talk about the rest of the program. Left-to-right evaluation is just a choice of  $K$ . No memory fact has to be mentioned. Facts that the hole does not use stay outside this rule and are framed.

**Lemma 4** (Frame).

$$P * \text{wp } e \{\Phi\} \vdash \text{wp } e \{v. P * \Phi(v)\}.$$

*Intuition.* If a proof of  $e$  never mentions  $P$ , then  $e$  cannot depend on the cells inside  $P$ , and those cells survive  $e$  unchanged. This is the rule that makes “the other pointer is untouched” a consequence of not putting that pointer into the callee’s precondition. It is not proved by induction on CJR expressions.

**Lemma 5** (Consequence).

$$\text{wp } e \{\Phi\} \wedge (\forall v. \Phi(v) \multimap \Psi(v)) \vdash \text{wp } e \{\Psi\}.$$

*Intuition.* A stronger postcondition can be weakened. The examples use this when a lemma such as `bump_wp` returns both a pure result and a framed cell, and the continuation only needs one of them. In Coq it is `wp_wand` or `wp_mono`.

### 5.4 Pure rules

A pure head step is one that is defined in every state, does not change the state, and has exactly one successor. The generic lemma is `wp_pure_step`. Each concrete rule below is an instance. Where the step produces a value immediately, the postcondition is under a later, because that is how the generic lemma returns the credit. Where the step produces another expression, the rule simply asks for a weakest precondition of that expression.

#### Closures and application.

```
Lemma wp_rec f x e Phi :
  later (pound 1 -*- Phi (RecV f x e)) |- wp (Rec f x e) Phi.
Lemma wp_app f x e v Phi :
  wp (subst_binder x v (subst_binder f (RecV f x e) e)) Phi
  |- wp (App (Val (RecV f x e)) (Val v)) Phi.
```

$$\begin{aligned} &\triangleright (\Phi(\text{rec } f \ x. e)) \vdash \text{wp } (\text{rec } f \ x. e) \{\Phi\}, \\ &\text{wp } e[x \mapsto v][f \mapsto \text{rec } f \ x. e] \{\Phi\} \vdash \text{wp } ((\text{rec } f \ x. e) \ v) \{\Phi\}. \end{aligned}$$

*Intuition.* Evaluating a function literal yields the closure as a value. Calling it substitutes the argument and, if the function is recursive, the closure itself. No heap fact is required: the code is data, and the data is the value in hand. That is what lets a vtable be an ordinary struct of closures.

Let, if, while, sequencing, arithmetic, offset.

$$\begin{aligned}
& \text{wp } e[x \mapsto v] \{ \Phi \} \vdash \text{wp } (\text{let } x = v \text{ in } e) \{ \Phi \} \\
& \text{wp } e_1 \{ \Phi \} \vdash \text{wp } (\text{if true then } e_1 \text{ else } e_2) \{ \Phi \} \\
& \text{wp } e_2 \{ \Phi \} \vdash \text{wp } (\text{if false then } e_1 \text{ else } e_2) \{ \Phi \} \\
& \text{wp } (\text{if } e_1 \text{ then } (e_2; \text{while } e_1 \text{ do } e_2) \text{ else } ()) \{ \Phi \} \vdash \text{wp } (\text{while } e_1 \text{ do } e_2) \{ \Phi \} \\
& \text{wp } e \{ \Phi \} \vdash \text{wp } (v; e) \{ \Phi \} \\
& \llbracket \oplus \rrbracket(v_1, v_2) = \text{Some}(v) \Rightarrow \triangleright(\Phi(v)) \vdash \text{wp } (v_1 \oplus v_2) \{ \Phi \} \\
& \triangleright(\Phi(\text{ptr}(\ell + i))) \vdash \text{wp } (\text{ptr}(\ell) + i) \{ \Phi \}.
\end{aligned}$$

*Intuition for while.* The loop rule does not mention an invariant. It unfolds the loop once into a conditional. The invariant is whatever assertion the user has in hand when they come back to the same syntactic loop. If the condition is false, the unfolding takes the else branch and returns unit. If it is true, the body runs and the same rule applies again. Termination is not proved; the unfolding is used only as many times as the concrete execution needs.

*Intuition for offset.* Adding an index to a pointer is arithmetic. It does not check that the resulting address lies in a block. The check happens when someone loads or stores: without a points-to fact at the computed address, the load rule does not apply, and the machine would be stuck. Bounds are therefore a matter of which points-to facts the proof owns, not of a side condition on  $+$ .

**Struct values.**

$$\begin{aligned}
& \triangleright(\Phi(\langle \bar{v} \rangle)) \vdash \text{wp } \langle \bar{v} \rangle \{ \Phi \}, \\
& \bar{v}(i) = v \Rightarrow \triangleright(\Phi(v)) \vdash \text{wp } (\langle \bar{v} \rangle.i) \{ \Phi \}.
\end{aligned}$$

*Intuition.* A finished zipper is the unboxed tuple. Reading a component copies it out. The tuple is not a heap cell, so there is nothing to frame and nothing to free. If the index is out of range, there is no rule.

Building a struct or an object from a non-empty expression list is also a pure step (**StructStepS**, **NewStepS**). The development uses `wp_pure_step` for that step in the vtable example rather than a named lemma.

## 5.5 Stack rules

```

Lemma wp_var_bind x v e Phi :
  (forall l, StackId l |->s v -*-
    wp (subst_var x l e)
      (fun w => exists v', StackId l |->s v' * Phi w))
  |- wp (VarBind x (Val v) e) Phi.

Lemma wp_stack_load l v Phi :
  l |->s v -*- later (pound 1 -*- l |->s v -*- Phi v)
  |- wp (StackLoad (stk l)) Phi.

Lemma wp_stack_assign l v w Phi :
  l |->s v -*- later (pound 1 -*- l |->s w -*- Phi unit)
  |- wp (StackAssign (stk l) (Val w)) Phi.

Lemma wp_stack_field l vs i w v Phi :

```

```

vs !! i = Some w ->
l |->s StructV vs -*-
  later (pound 1 -*- l |->s StructV (set_nth i v vs) -*- Phi unit)
  |- wp (StackFieldStore (stk l) i (Val v)) Phi.

```

In mathematics, with  $\ell$  a stack address:

$$\begin{aligned}
& (\forall \ell. \ell \mapsto_s v \multimap \text{wp } e \{x \mapsto \ell\} \{w. \exists v'. \ell \mapsto_s v' \multimap \Phi(w)\}) \vdash \text{wp } (\text{var } x = v \text{ in } e) \{\Phi\}, \\
& \ell \mapsto_s v \multimap \triangleright (\ell \mapsto_s v \multimap \Phi(v)) \vdash \text{wp } (\text{stk}(\ell)) \{\Phi\}, \\
& \ell \mapsto_s v \multimap \triangleright (\ell \mapsto_s w \multimap \Phi(())) \vdash \text{wp } (\text{stk}(\ell) := w) \{\Phi\}, \\
& \bar{v}(i) = w \Rightarrow \ell \mapsto_s \langle \bar{v} \rangle \multimap \triangleright (\ell \mapsto_s \langle \bar{v}[i \mapsto u] \rangle \multimap \Phi(())) \vdash \text{wp } (\text{stk}(\ell).i := u) \{\Phi\}.
\end{aligned}$$

*Intuition for **var**.* The rule allocates a fresh slot, hands you exclusive ownership of it, and asks you to prove the body after the name has been compiled into stack operations. You may change the contents. At the end you must give the slot back, at whatever value it then has. The **pop** administrative step deletes it, so the caller does not see a stack fact. This is lexical scope: the owner of the cell is the body, and the cell does not escape.

The existential is essential. If the postcondition demanded the original value, an assignment inside the body could not be verified, because assignment replaces the points-to fact.

*Intuition for **load**, **assign**, and **field update**.* Load reads and returns the same fact. Assignment exchanges the fact for one about the new value. Field update is assignment of a struct, with one component changed, and it is only available when that component already exists. A missing component is stuck, and there is no points-to fact that would let the rule fire. None of these rules allocates on the raw heap or the object heap.

## 5.6 Raw-heap rules

```

Lemma wp_alloc_one Phi :
  (forall l, RawId l |->r LitV (LitInt 0) -*-
    RawId l |->b 1 -*- Phi (LitV (LitPtr l)))
  |- wp (Alloc (Val (LitV (LitInt 1)))) Phi.

Lemma wp_free_one l v Phi :
  RawId l |->r v -*- RawId l |->b 1 -*- later (Phi unit)
  |- wp (Free (Val (LitV (LitPtr l)))) Phi.

Lemma wp_load l v Phi :
  RawId l |->r v -*- later (RawId l |->r v -*- Phi v)
  |- wp (Load (Val (LitV (LitPtr l)))) Phi.

Lemma wp_store l v w Phi :
  RawId l |->r v -*- later (RawId l |->r w -*- Phi unit)
  |- wp (Store (Val (LitV (LitPtr l))) (Val w)) Phi.

```

$$\begin{aligned}
& (\forall \ell. \ell \mapsto_r 0 \multimap \ell \mapsto_b 1 \multimap \Phi(\text{ptr}(\ell))) \vdash \text{wp } (\text{alloc}(1)) \{\Phi\}, \\
& \ell \mapsto_r v \multimap \ell \mapsto_b 1 \multimap \triangleright (\Phi(())) \vdash \text{wp } (\text{free}(\ell)) \{\Phi\}, \\
& \ell \mapsto_r v \multimap \triangleright (\ell \mapsto_r v \multimap \Phi(v)) \vdash \text{wp } (!\ell) \{\Phi\}, \\
& \ell \mapsto_r v \multimap \triangleright (\ell \mapsto_r w \multimap \Phi(())) \vdash \text{wp } (\ell := w) \{\Phi\}.
\end{aligned}$$

*Intuition.* Allocation creates a cell you did not own before, initialised to zero, and a separate fact that a block of length one starts there. The two facts have different jobs. The points-to fact lets you load and store the word. The block fact lets you free the block. Free consumes both, so the word cannot be used again: there is no fact left to feed to the load rule, and the concrete word has been deleted, so the machine would get stuck if you tried.

Load does not disturb the cell. Store keeps ownership and changes the value. Neither rule needs the block fact. A client may therefore load and store while the right to free sits unused in the context, and the frame rule keeps it there.

The operational rule for allocation accepts any positive length. The verified rule above is the one-word instance, which is what the examples use. A longer block would be the same pattern: a points-to fact per word, plus one block fact for the base and the length.

## 5.7 Object rules

```

Lemma wp_new vs Phi :
  (forall o, ([* list] i |-> v in vs, ObjId o |->o[i] v) -*-
    Phi (LitV (LitObj o)))
  |- wp (New vs []) Phi.

Lemma wp_field_load o i v Phi :
  ObjId o |->o[i] v -*- later (ObjId o |->o[i] v -*- Phi v)
  |- wp (FieldLoad (Val (LitV (LitObj o))) i) Phi.

Lemma wp_field_store o i v w Phi :
  ObjId o |->o[i] v -*- later (ObjId o |->o[i] w -*- Phi unit)
  |- wp (FieldStore (Val (LitV (LitObj o))) i (Val w)) Phi.

```

$$\begin{aligned}
& (\forall o. \bigstar_{i < k} o \mapsto_o [i]v_i \multimap \Phi(\text{obj}(o))) \vdash \text{wp}(\text{new}(v_0, \dots, v_{k-1})) \{\Phi\}, \\
& o \mapsto_o [i]v \multimap \triangleright(o \mapsto_o [i]v \multimap \Phi(v)) \vdash \text{wp}(o.i) \{\Phi\}, \\
& o \mapsto_o [i]v \multimap \triangleright(o \mapsto_o [i]w \multimap \Phi(())) \vdash \text{wp}(o.i := w) \{\Phi\}.
\end{aligned}$$

*Intuition.* `new` of a finished argument list allocates a fresh object and returns one points-to fact per field, with the initialising values. There is no block fact and no free rule: the object stays allocated. Field load and field store are the same idea as raw load and store, on a different map. Owning  $o \mapsto_o [i]v$  says nothing about  $o \mapsto_o [j]w$  for  $j \neq i$ , and nothing about any raw word. That is why a raw cell can be framed across a field update, and why two fields of one object can be reasoned about separately.

## 5.8 Exclusivity

```

Lemma stack_mapsto_exclusive l v1 v2 :
  l |->s v1 -*- l |->s v2 |- False.
Lemma raw_mapsto_exclusive l v1 v2 :
  l |->r v1 -*- l |->r v2 |- False.

```

**Lemma 6** (Exclusive ownership). *For a stack address or a raw address  $\ell$ , and any values  $v_1, v_2$ ,*

$$\ell \mapsto_s v_1 * \ell \mapsto_s v_2 \vdash \perp, \quad \ell \mapsto_r v_1 * \ell \mapsto_r v_2 \vdash \perp.$$

*Intuition.* Full ownership is unique. The ghost-map algebra rejects two fragments of fraction 1 at the same key, even if they agree on the value. This is the specification of non-aliasing. If a swap is specified with two points-to assumptions, applying it to one address twice is impossible: the precondition is false. The machine might still have a single cell there; the logic refuses to pretend there are two.

The same algebra applies to block facts and to object fields. The development records the stack and raw cases because those are the ones the examples need to point at.

## 6 Main theorems

### 6.1 Head reduction is deterministic

Theorem 1 above is `base_step_det`. Restated: the head reduction relation is a partial function of  $(e, \sigma)$ . Combined with the context machinery, a configuration has at most one successor. Stuckness is therefore unambiguous. Either the term is a value, or exactly one head redex is ready and the rule applies, or no rule applies and the program is stuck.

### 6.2 Well-formedness is an invariant of the verified steps

The state interpretation contains  $\ulcorner \text{wf}(\sigma) \urcorner$ . Each memory rule proves that the state after the step is well formed if the state before was. The statements are `wf_stack_alloc`, `wf_stack_delete`, `wf_stack_upd`, `wf_raw_store`, `wf_alloc_one`, `wf_free_one`, `wf_new`, and `wf_field_store`.

**Lemma 7** (Allocator invariant).  $\text{wf}(\sigma_0)$ . *If  $\text{wf}(\sigma)$  and a verified memory rule steps to  $\sigma'$ , then  $\text{wf}(\sigma')$ .*

The one-word cases are the ones used by the logic. For example, freeing a block of length 1 at  $\ell$  preserves coverage of every other block because disjointness puts their words outside  $\{\ell\}$ .

### 6.3 Adequacy

```
Lemma cjr_adequacy S e (phi : val -> Prop) :
  (forall '!cjrGS S} '!invGS_gen HasLc S},
    |- wp e (fun v => pure (phi v))) ->
    adequate NotStuck e state_init (fun v _ => phi v).
```

**Theorem 8** (Adequacy). *Let  $\varphi$  be a pure predicate on values. Suppose that in every Iris instantiation of the CJR ghost state one can prove*

$$\text{wp } e \{ v. \ulcorner \varphi(v) \urcorner \}.$$

*Then, executing  $e$  from  $\sigma_0$ , every reachable configuration can take a step or is a value, and every final value  $v$  satisfies  $\varphi(v)$ .*

*What the proof does.* It allocates four empty ghost maps, packages them as the CJR ghost state, and observes  $\text{wf}(\sigma_0)$ . Iris’s adequacy theorem for weakest preconditions then applies. The hypothesis is a proof that does not depend on a particular ghost name, so it can be used at the names chosen here.

*What it does not say.* It does not say that every closed expression is safe. An expression with a free variable, a null dereference, or a load of an address nobody allocated has no such proof, and the

theorem is silent. It also does not extract a functional-correctness result beyond the pure  $\varphi$  that was proved. If the postcondition is only “the result is unit”, adequacy says the program returns unit without getting stuck; the interesting memory facts were used and consumed inside the proof.

The closed examples `sum_one`, `bump_keep`, `cell`, `class_bump`, and `vtable_prog` are instantiated at this theorem. The other examples are open: their hypotheses are points-to facts, so they are stated as weakest preconditions rather than as runs from  $\sigma_0$ .

## 7 The examples

Each example uses a feature the previous one does not. They are the files in `theories/examples/`, in the order below. The programs are written here in surface syntax; the Coq definitions are the `expr` terms those files actually prove.

### 7.1 Arithmetic: `arith.v`

**Program.**

```
var n = 1 in
var s = 0 in
while (1 ≤ n) do (s := s + 1; n := n - 1);
s.
```

**Theorem.**

$$\text{wp sum\_one} \{ v. \ulcorner v = 1 \urcorner \}.$$

From  $\sigma_0$ , the program returns 1 and does not get stuck (`sum_one_adequate`).

**What is new.** Nothing is allocated on either heap. The proof uses `var`, stack load and assignment, `while`, `if`, sequencing, and integer  $+$  and  $-$ . The loop condition is true once:  $n$  is 1, then 0. On the true branch,  $s$  becomes 1 and  $n$  becomes 0. On the false branch the loop returns unit, and the trailing read returns  $s$ .

The stack facts at the end are  $s \mapsto_s 1$  and  $n \mapsto_s 0$ . They are given back to `wp_var_bind` through the existential, and `pop` deletes them. The pure postcondition does not mention the stack. That is the scope rule from the previous section, exercised on a loop that really mutates.

### 7.2 Stack structs: `struct_upd.v`

**Program.**

```
var p = ⟨0⟩ in
(p.0 := 1; p.0).
```

A second, closed program allocates a raw word and then runs the first program, then loads the raw word:

```
let r = alloc(1) in (bump; !r).
```

**Theorems.**

$$\begin{aligned} \ell \mapsto_r v \vdash \text{wp bump} \{ w. \lceil w = 1 \rceil * \ell \mapsto_r v \}, \\ \text{wp bump\_keep} \{ v. \lceil v = 0 \rceil \}. \end{aligned}$$

The closed run from  $\sigma_0$  returns 0 (`bump_adequate`).

**What is new.** The struct lives in a stack slot, not on the raw heap. The field write is `wp_stack_field`: the slot changes from  $\langle 0 \rangle$  to  $\langle 1 \rangle$ , and the read copies the component out with `wp_struct_load`. No raw address is allocated by `bump` itself.

The first theorem is the frame. Its hypothesis is an arbitrary raw cell, and the postcondition returns that same cell. The proof of the stack update never mentions it, so it cannot have been changed. The closed program makes the same point operationally: the fresh raw word is 0 before `bump` and 0 after, which is what the final load returns. The field update’s result 1 is established on the way and then discarded by sequencing, because the observation we keep is the unchanged raw word.

**7.3 One raw cell: `cell.v`****Program.**

$$\text{let } p = \text{alloc}(1) \text{ in } (p := 7; (!p; \text{free } p)).$$

**Theorem.**

$$\text{wp cell} \{ v. \lceil v = () \rceil \}.$$

From  $\sigma_0$ , the program returns unit and does not get stuck.

**What is new.** This is the first use of  $\mapsto_r$  and  $\mapsto_b$ . Allocation produces  $\ell \mapsto_r 0$  and  $\ell \mapsto_b 1$ . The store replaces the points-to fact by  $\ell \mapsto_r 7$ . The load reads 7 and returns the same fact. The program then discards that 7 by sequencing and frees the block. Free consumes both facts and returns unit.

The pure specification does not say “the loaded value was 7”. That fact is internal to the proof. What adequacy reports is that the whole expression, whose value is the value of `free`, returns unit without a stuck load, store, or double free. After `free`, no points-to fact remains, so a further load could not be verified.

**7.4 Swap and aliasing: `swap.v`**

**Program.** For addresses  $\ell_1$  and  $\ell_2$ ,

$$\text{let } t = !\ell_1 \text{ in } (\ell_1 := !\ell_2; \ell_2 := t).$$

**Theorems.**

$$\begin{aligned} \ell_1 \mapsto_r v_1 * \ell_2 \mapsto_r v_2 \vdash \text{wp swap}(\ell_1, \ell_2) \{ \neg. \ell_1 \mapsto_r v_2 * \ell_2 \mapsto_r v_1 \}, \\ \ell \mapsto_r v_1 * \ell \mapsto_r v_2 \vdash \perp. \end{aligned}$$

**What is new.** The postcondition needs both cells. The proof loads  $\ell_2$  while still owning  $\ell_1$ , stores into  $\ell_1$ , then stores the saved value into  $\ell_2$ . Each rule mentions only the cell it touches. The other fact sits in the context and is framed.

The second lemma is `raw_mapsto_exclusive`, re-exported as `swap_aliased`. It is not a second proof of the same swap. It says that the swap specification cannot be instantiated at one address twice: the hypothesis of two points-to facts at that address is absurd. An overlapping specification would be a different theorem, for example a single cell updated twice, and it is not claimed here.

## 7.5 Indexed blocks: `sum_block.v`

**Program.** Given a pointer  $p$ ,

```
var i = 0 in
var s = 0 in
while (i ≤ 0) do
  (s := s + !(p + i); i := i + 1);
s.
```

**Theorem.**

$$p \mapsto_r 4 * p \mapsto_b 1 \vdash \text{wp sum\_at}(p) \{ v. \ulcorner v = 4 \urcorner * p \mapsto_r 4 * p \mapsto_b 1 \}.$$

**What is new.** The address that is loaded is not the base written in the source. It is the value of `Offset`, that is  $p + i$ . On the only iteration,  $i$  is 0, and  $p + 0 = p$ , so the load is justified by the points-to fact at  $p$ . The addition itself is the pure offset rule; it does not consume the points-to fact. The loop then increments  $i$  to 1, the condition  $i \leq 0$  fails, and the sum 4 is returned.

The block-length fact is never used by the load, and it is still there at the end. That is the split between “I may read this word” and “I may free this block”. The example does not free. A load of  $p + 1$  would not be provable from these hypotheses: there is no points-to fact at  $p + 1$ , and a one-word block does not contain that word.

The loop invariant, in the ordinary sense, is the assertion carried across the unfolding:  $i$  and  $s$  are stack cells,  $s$  holds the sum of the prefix already visited, and the raw word is unchanged. For a one-iteration loop the prefix is either empty or the single element 4.

## 7.6 Final classes: `class_upd.v`

**Program.**

```
let r = alloc(1) in
let o = new(0) in
(o.0 := 5; (o.0; !r)).
```

**Theorem.**

$$\text{wp class\_bump} \{ v. \ulcorner v = 0 \urcorner \}.$$

From  $\sigma_0$ , the program returns 0 and does not get stuck.

**What is new.** `new` allocates on the object map, not the raw map. The client receives  $o \mapsto_o [0]0$ . The field store changes it to  $o \mapsto_o [0]5$ , and the field load reads 5. The raw word allocated first is 0, is never stored, and is what the program returns.

So the closed observation is again the framed cell: object update does not change the raw heap. The proof makes the separation concrete. After `wp_alloc_one`, the raw points-to fact stays in the context through `wp_new`, `wp_field_store`, and `wp_field_load`. None of those lemmas takes a raw fact as a hypothesis. The final `wp_load` is what consumes it.

## 7.7 Framing across a call: `frame_call.v`

**Program.** For pointers  $p$  and  $q$ ,

$$\begin{aligned} &\text{let } f = (\text{fun } x. x := 9) \text{ in} \\ &(f\ p; !q). \end{aligned}$$

**Theorem.**

$$p \mapsto_r v * q \mapsto_r 3 \vdash \text{wp write\_call}(p, q) \{ w. \ulcorner w = 3 \urcorner * p \mapsto_r 9 * q \mapsto_r 3 \}.$$

**What is new.** The callee is a closure. `wp_rec` makes the closure, `wp_app` substitutes the pointer  $p$  for  $x$ , and `wp_store` needs only  $p \mapsto_r v$ . The fact  $q \mapsto_r 3$  is not in the callee’s precondition. It is framed across the call, then used by the load, which returns 3.

This is the frame rule applied at a function boundary. Nothing about `App` is special. Substitution puts a value in the body, and the body is an ordinary store. The guarantee that  $q$  is unchanged is the guarantee that the store’s proof did not receive  $q$ .

## 7.8 A hand-written `vtable`: `vtable.v`

**Program.**

$$\begin{aligned} &\text{let } inc = (\text{fun } this. this.0 := this.0 + 1) \text{ in} \\ &\text{let } box = \langle inc \rangle \text{ in} \\ &\text{let } o = \text{new}(0) \text{ in} \\ &(box.0) o; o.0. \end{aligned}$$

**Theorem.**

$$\text{wp vtable\_prog} \{ v. \ulcorner v = 1 \urcorner \}.$$

From  $\sigma_0$ , the program returns 1 and does not get stuck.

**What is new.** The function is a value, the struct is a tuple holding that value, and the call is an ordinary application of a component that was loaded out of the tuple. There is no method table in the machine and no dynamic dispatch rule. A “method” is a function whose first argument is the object, and a “vtable” is a struct the programmer assembled.

The proof follows that reading. The closure is formed by `wp_rec`. The struct zipper evaluates the closure and becomes  $\langle inc \rangle$  by the pure struct rules. `new` yields  $o \mapsto_o [0]0$ . `wp_struct_load` copies the closure out; this does not touch the object. `wp_app` substitutes the object reference into the body. The body loads field 0, adds one, and stores 1. The trailing field load reads that 1, which is the value of the whole program.

All three regions of the story meet: a stack-free unboxed struct of code, a code pointer, and an object field. The raw heap is not used. The earlier examples already showed that leaving it unused is safe.

## 8 What is deliberately missing

The semantics does not include concurrency, so there is no invariant-opening rule and no fork. Float literals exist, and no float operator does. Allocation in the machine accepts a positive word count; the named logic rule used throughout the examples is the one-word rule. Objects are not collected. Pointer arithmetic is word-addressed. Null, a freed address, and an address with no points-to fact are stuck rather than given a defined error value.

The type grammar is not enforced. The adequacy theorem is the safety theorem for those programs whose specifications have been proved.